

the given problem. For example, a lot of training needs to be done to build the neural network of data sets to get greater accuracy in recognising images.

Computational problems are largely solved using a traditional approach like statistical and scientific algorithms. However, some of these problems prove to be costly in terms of time and effort if solved in this way. Pretrained machine learning models can handle such problems using less time and money.

Vision API, image API and face detection

AI grew in importance from the early 1950s to the late 1980s, when engineers developed intelligent machines and programs to address particular problems.

Since training AI models was costly and time consuming, and the accuracy rate or efficiency of the AI algorithms remained static, ML evolved in the 1980s as a subset of AI. ML models learn by themselves without being explicitly programmed, as a self-training capability is slowly built into them. They address specific problems and can be used for image or speech recognition with higher accuracy.

Deep learning (DL) is a subset of ML, which uses a multi-layered neural network and improves accuracy and efficiency by learning from its experience (called historical data). It builds a model based on the history, and is very efficient for large volumes of data. Deep learning started to emerge in the late 2010s and is widely used in speech recognition, image or pattern recognition, object detection in videos and images, and language translation, to name a few.

Cognitive services help to quickly develop AI models for applications to manage human-like decision making services such as pattern recognition, speech recognition and search facilities. Machine learning cognitive services can be grouped into five major categories of

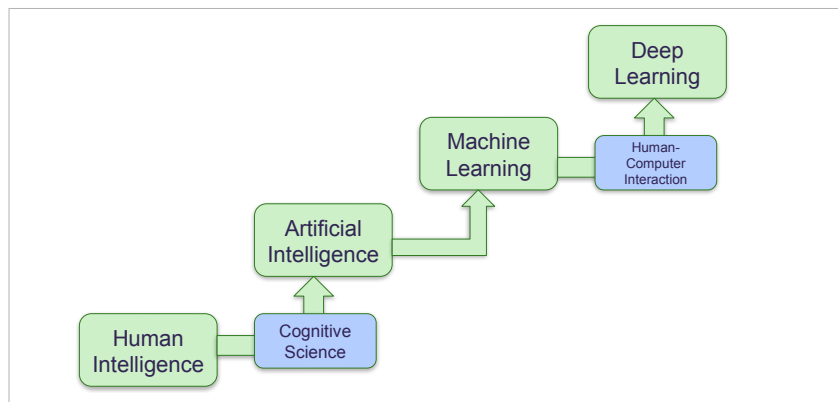


Figure 2: Cognitive intelligence types

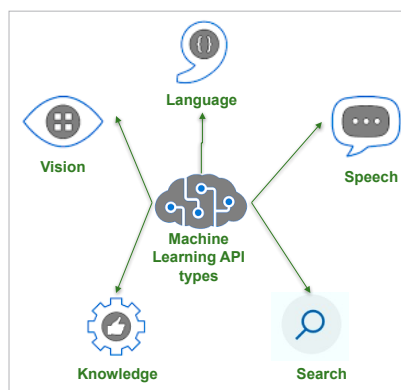


Figure 3: Machine learning API types

activities, as given below.

Vision API: This service helps to handle image and video processing in face detection, object detection, and pattern recognition with specific use case features like celebrity/landmark detection, handwritten character recognition and OCR activities.

Language API: This service handles sentiment analysis, language understanding (termed as LUIS), bot facilities for answering questions from the knowledge base, language detection, language translation, phonetic analysis, etc.

Speech API: This service handles speech-to-text and text-to-speech translation, real-time speech translation and transcribing, custom voice moderation, etc.

Knowledge API: This service facilitates bot services, contact centre AI facilities for knowledge base creation, QA

extraction from textual documentation, self-training from voice answers, content customisation, personalised learning, semantic matching, anomaly detection in content search, etc.

Search API: Based on Bing or Google search APIs, this service offers a search facility for images, videos, text and audio content; image identification, extraction and classification; lead generation based on search results, etc.

OpenCV

OpenCV is one of the most popular vision API facilitated open source machine learning frameworks. It provides various libraries and hardware simulators for building, deploying and testing machine learning models. One of the most important implementers of OpenCV is the Microsoft Azure platform, which uses it for its MMLSpark framework for advanced Big Data services.

Microsoft has contributed to the Apache Spark framework by introducing the Microsoft Machine Learning for Apache Spark (MMLSpark) framework. MMLSpark expands the distributed machine learning solution of Apache Spark by adding OpenCV, Microsoft Cognitive Toolkit (CNTK) and many other machine learning framework capabilities to the Spark framework in Big Data processing using the enterprise data lake solution.

MMLSpark comprises a group of machine learning libraries. These are listed below.